

- Order of Reaction
- Reverse Osmosis
- Miers' theory in crystallization

5th Sem/ Chemical (Pulp & Paper)
Subject : Chemical Engineering Operations

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Distillation separates components based on differences in :
- a) Color
 - b) Density
 - c) Boiling point
 - d) Melting point
- Q.2 Azeotropic distillation is used to separate:
- a) Immiscible liquids
 - b) Ideal mixtures
 - c) Azeotropes
 - d) Gases
- Q.3 Which condition helps in crystal formation?
- a) Heating
 - b) Super saturation
 - c) Stirring
 - d) Cooling fast
- Q.4 Dialysis is used in:
- a) Boilers
 - b) Reactors
 - c) Kidney treatment
 - d) Crystallizers

- Q.5 Enthalpy is a measure of:
 a) Work done b) Heat content
 c) Force d) Mass
- Q.6 Chemical reaction that proceeds in both directions is called:
 a) Irreversible b) Reversible
 c) Parallel d) Complex

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Expand VLE.
 Q.8 Which theory explains crystall growth?
 Q.9 What is the driving force in reverse osmosis?
 Q.10 Name any one membrane separation processes.
 Q.11 What is the full form of PFR.
 Q.12 _____ ratio effects the number of theoretical stages in distillation column.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Define relative volatility and its significance in distillation.

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- Q.14 Define crystallization and list its industrial applications.
 Q.15 Define supersaturation and explain how it affects crystallization.
 Q.16 Differentiate between microfiltration and ultrafiltration.
 Q.17 Define internal energy, enthalpy and entropy.
 Q.18 Differentiate between open, close and isolated systems.
 Q.19 Write a short note on dialysis with an example.
 Q.20 Compare reversible and irreversible reaction with examples.
 Q.21 Discuss various types of distillation techniques.
 Q.22 Explain the difference between elementary and non-elementary reactions.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the McCabe-Thiele method with the help of a neat diagram and step-by-step procedure.
 Q.24 Classify different types of crystallizers and briefly explain any one with the help of neat diagram.

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